

Quarry AI

AI Competitive Intelligence Agent

RAG + Multi-Agent System • Product Requirements Document

Portfolio Project • AI Product Manager

What this product does

Quarry AI is an always-on AI analyst that continuously monitors competitors, retrieves real-time signals from websites, job boards, news, and reviews, then answers strategic questions with cited, grounded answers — in under 5 seconds.

A. Problem Statement

The pain in one sentence

Product teams spend 6–10 hours per week manually tracking competitors. By the time they find a signal, the decision has already been made.

What the current process looks like

- Slow — manual research takes days; competitor changes happen in hours
- Fragmented — each team member tracks different sources with no shared truth
- Non-scalable — the problem gets worse as the competitor set grows
- Prone to missed signals — a competitor can launch a major feature on Monday and the team finds out Friday
- No audit trail — no record of what competitor data informed past pricing or roadmap decisions

The cost of missing a signal

A sales team learns about a competitor pricing change from a prospect mid-call, not from internal intelligence. A PM deprioritises a feature only to watch a competitor ship it and gain market share. These are not edge cases — they are weekly occurrences in fast-moving markets.

Goal

PM goal (not engineering goal)

Build an AI agent that automatically monitors competitors and delivers actionable strategic insights — cited, fresh, and fast enough to use in a board meeting.

B. User Persona

Primary users

Sarah — Director of Product

Goal: Understand what competitors are building to differentiate the roadmap.

Frustration: Team wastes 3 hrs/week on manual checks. Updates are always late.

Uses Quarry AI for: Weekly competitor briefings before roadmap reviews.

Marcus — Head of Sales

Goal: Know competitor pricing and messaging before entering a deal.

Frustration: Learns about competitor pricing changes from prospects, not from internal intel.

Uses Quarry AI for: Pre-call competitor summaries and battle card updates.

Shared pain points

- Recency problem — both need current information, not last week's snapshot
- Trust problem — neither trusts AI summaries without knowing the source; citations are non-negotiable
- Speed problem — both are in back-to-back meetings; answers must arrive in under 5 seconds

C. System Architecture

PM framing

The architecture has four distinct layers so each can fail, be monitored, and be improved independently. If retrieval quality drops, I know exactly which layer to fix without touching the others.

Layer 1 — Data ingestion (always running in background)

- Competitor websites — pricing pages, product pages (crawled daily)
- News APIs — TechCrunch, PR Newswire (live RSS)
- App store reviews — G2, Capterra, Apple/Google stores (daily)
- Crunchbase / SEC filings — funding data, 10-K/10-Q for public competitors (weekly)
- Reddit / forums — community sentiment signals (live)
- Job postings — LinkedIn, Greenhouse (daily) — hiring patterns reveal strategic intent

Layer 2 — Multi-agent orchestration

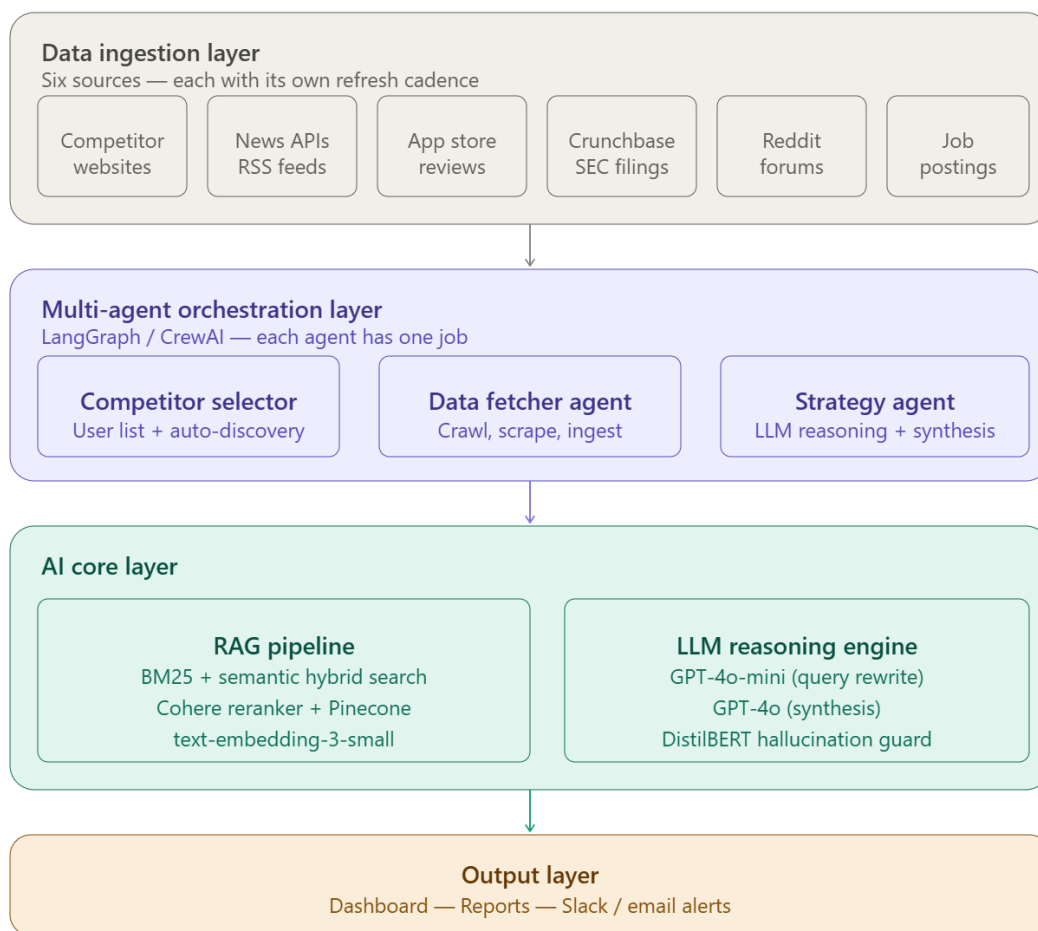
- Competitor selector agent — user-defined list + automatic discovery by industry category
- Data fetcher agent — crawl, scrape, and ingest from all source types
- Strategy agent — LLM reasoning layer that synthesises insights from retrieved context
- Memory store (Redis) — tracks what was already retrieved and what changed since last crawl

Layer 3 — AI core (RAG pipeline)

- Chunking — split and deduplicate text from all sources
- Embeddings — text-embedding-3-small (not -large; 5x cheaper, sufficient for this use case)
- Vector store — Pinecone / Weaviate
- Hybrid retrieval — BM25 for exact product names + semantic search for intent matching
- Reranker — Cohere Rerank as a second retrieval stage to score chunk relevance before synthesis
- LLM synthesis — GPT-4o for final answer generation with strict citation enforcement

Layer 4 — Output

- Reports — structured competitor summaries with cited sources
- Dashboard — competitor cards with freshness indicators and alert badges
- Alerts — Slack / email notifications when a monitored signal changes



Purple = agent layer Teal = AI core Amber = output

D. System Flow

Two parallel pipelines

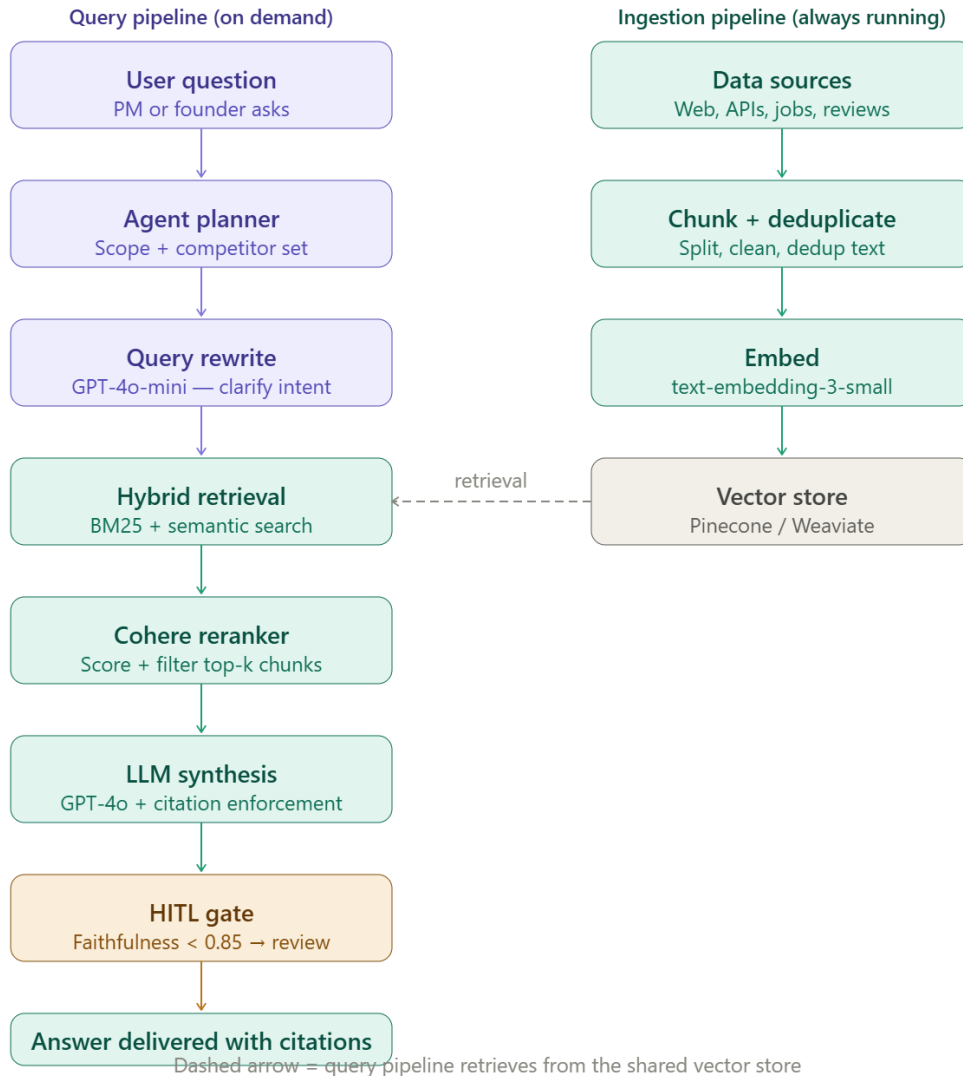
The ingestion pipeline runs continuously in the background — it optimises for freshness, not speed. The query pipeline activates on each user question and must respond in under 5 seconds. Separating them means each can be optimised for its own constraint.

Query pipeline (on demand)

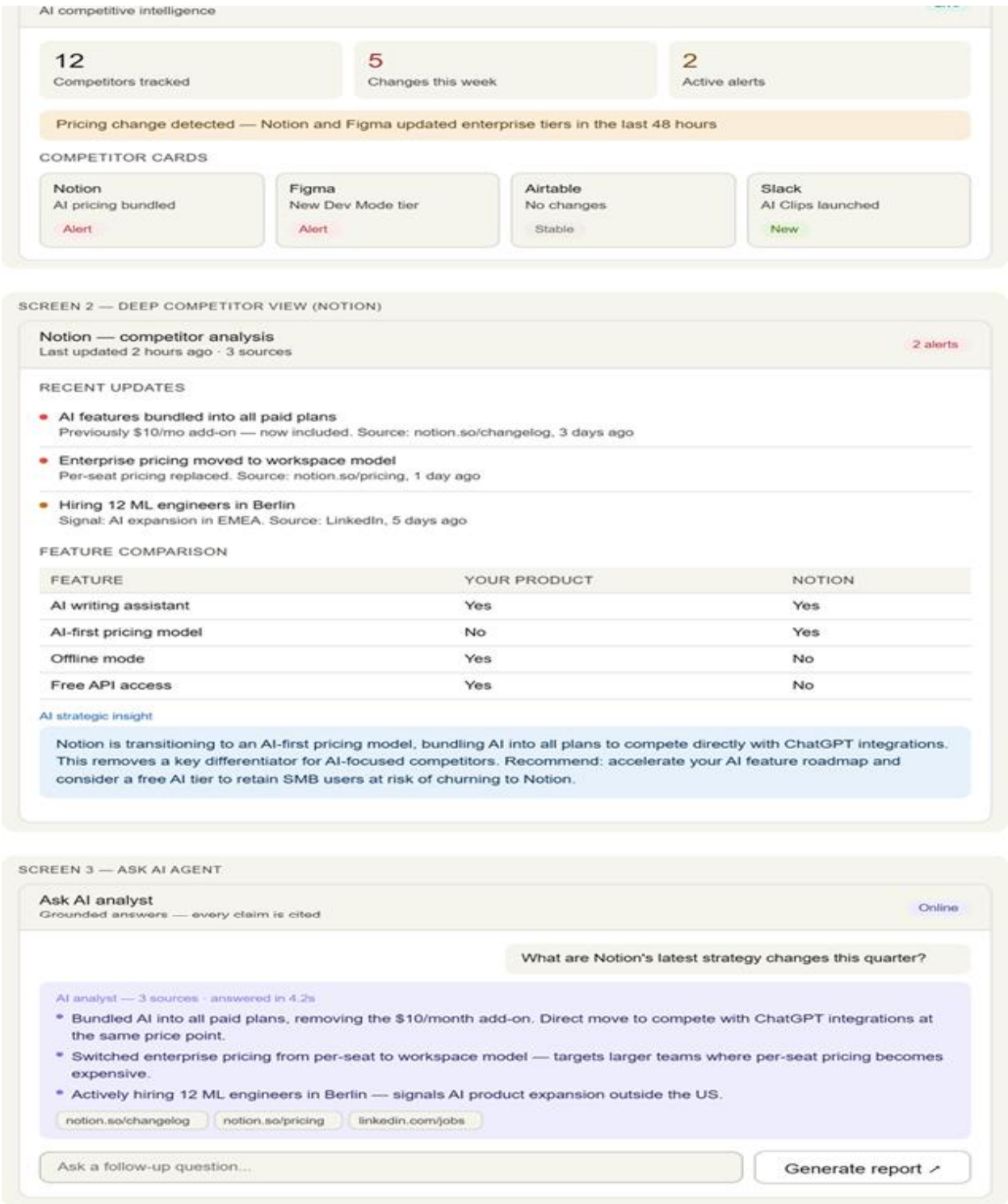
- User submits question (e.g. “What is Notion doing in AI this quarter?”)
- Agent planner decides scope, competitor set, and which agents to activate
- Query rewrite agent (GPT-4o-mini) clarifies and expands the query for better retrieval
- Retrieval agent performs hybrid search (BM25 + semantic) against the vector store
- Reranker scores and filters the top-k chunks for relevance
- Synthesis agent (GPT-4o) generates a cited answer from retrieved context only
- HITL gate — if faithfulness score < 0.85, answer is routed for human review before delivery
- Answer delivered with source URLs, timestamps, and freshness indicator

Ingestion pipeline (background)

- Crawlers run on schedule per source (daily/hourly/weekly)
- Text is chunked, deduplicated, and embedded
- Chunks are upserted into the vector store with source URL and timestamp metadata
- Stale chunks (> 14 days) are flagged and re-fetched on next crawl cycle



E. UI DESIGN



F. Model Stack & Technical Choices

Agent / step	Model chosen	PM rationale
Chunking + crawling	No LLM	Pure Python. LLMs add cost with zero benefit at this stage.
Embedding	text-embedding-3-small	5x cheaper than -large (\$0.002 vs \$0.013 per 1M tokens). Sufficient quality for competitor text retrieval.
Query rewrite	GPT-4o-mini	Fast and cheap. Clarifies vague queries before retrieval. Does not require reasoning depth.
Retrieval	Pinecone + Cohere Rerank	Hybrid BM25 + vector search catches exact product names (BM25) and semantic intent (vector). Reranker adds relevance scoring.
Synthesis	GPT-4o	Reserved for the final step only where reasoning quality and citation accuracy are critical.
Hallucination guard	DistilBERT classifier	Fine-tuned to check if each claim is grounded in retrieved chunks. Runs post-synthesis before delivery.

G. Data Sources

Six source types with refresh cadence

- Competitor websites — pricing pages, product pages (crawled daily via Playwright stealth mode)
- News APIs — TechCrunch, PR Newswire, G2 review deltas (live RSS)
- App store reviews — Apple App Store, Google Play, G2, Capterra (daily)
- Public datasets — Crunchbase funding, SEC 10-K / 10-Q filings for public competitors (weekly)
- Social signals — Reddit, Hacker News mentions, community sentiment (live)
- Job postings — LinkedIn, Greenhouse, Lever (daily) — hiring patterns signal strategic intent 6–9 months ahead

Why job postings matter as a data source

A competitor hiring 10 ML engineers signals an AI product launch 6–9 months in advance. This is a strategic intelligence signal that most competitive tools miss entirely. It is a PM decision to include it, not an engineering default.

H. Evaluation Metrics

Three-tier evaluation framework

Retrieval quality tells you if the right chunks are found. Generation quality tells you if the AI used them correctly. Product metrics tell you if the PM got something they could act on. All three tiers must be healthy for the product to work.

Tier 1 — Retrieval quality (automated nightly via RAGAS framework)

Metric	Target	What it measures
Context precision	> 80%	Of retrieved chunks, how many were actually relevant to the query
Context recall	> 75%	Of all needed context, how much was successfully retrieved
Noise sensitivity	< 20%	How much irrelevant content degrades the final answer quality

Tier 2 — Generation quality (automated nightly)

Metric	Target	What it measures
Faithfulness	> 90%	Every claim in the answer is grounded in a retrieved chunk
Response relevance	> 85%	The answer addresses what the user actually asked
Factual correctness	> 85%	Facts in the answer are accurate and verifiable
Hallucination rate	< 5%	Frequency of fabricated or unsupported content in answers

Tier 3 — Product metrics (measured weekly)

Metric	Target	Definition
Task Success Rate	> 80%	User marks answer useful OR does not submit a follow-up within 60 seconds. Primary north star metric.
Citation click-through	> 30%	Proxy for trust. If users never click sources, they do not trust answers.
Query abandonment rate	< 20%	Users who ask a follow-up within 30 seconds — signal the answer was insufficient.
Report adoption rate	> 60%	Reports that are used in an actual meeting or decision within 7 days of generation.

Why Task Success Rate, not accuracy

A technically accurate answer that a PM cannot act on has failed. Task Success Rate measures the product outcome, not the model output. This is the difference between a model metric and a product metric.

I. Latency & Cost

Latency budget per query

Step	Time	Note
Query rewrite (GPT-4o-mini)	~0.4s	Fast model, short prompt, single output
Vector retrieval (Pinecone)	~0.3s	P95 at < 10M vectors
Cohere reranker	~0.3s	Reranks top-8 chunks for relevance
GPT-4o synthesis (streamed)	~3.2s	Mitigated by streaming — first tokens appear in < 1s
Hallucination guard	~0.2s	DistilBERT runs in parallel with streaming
Total P95	< 5s	Meets the constraint: users are in meetings

Cost per query breakdown

- GPT-4o-mini (query rewrite) — ~\$0.002
- GPT-4o (synthesis, ~3,000 token context) — ~\$0.016
- Cohere reranker — ~\$0.002
- Pinecone reads — ~\$0.004
- Total per query — ~\$0.024 (under the \$0.03 target)

Cost-per-seat business math

At ~\$0.024/query and 10 queries/user/day = \$0.24/user/day = ~\$7.20/user/month. At \$49/seat pricing, gross margin on compute is healthy. This calculation makes the business case concrete, not aspirational.

J. Failure Modes & Mitigations

Failure mode	Severity	Mitigation
Hallucinated competitor features	High	Citation enforcement in system prompt + faithfulness classifier. Answers below 0.85 confidence routed to HITL before delivery.
Outdated pricing data	High	Freshness timestamp on every answer. Staleness warning if best chunk > 7 days old. Daily crawl cadence for pricing pages.
Crawler blocked by anti-bot protection	Medium	Playwright stealth mode + rotating proxies. Fallback to Google Cache. Alert fires when crawl failure rate > 20% for any competitor.
Poor retrieval relevance (wrong chunks)	Medium	Cohere reranker as second retrieval stage. Context precision monitored daily. PagerDuty alert if precision drops below 70%.
Over-generalised insights	Medium	Synthesis prompt enforces specificity: 'Do not state market trends. State only what this competitor did, when, and what it signals.'
GPT-4o API latency spike	Low	Automatic fallback to GPT-4o-mini if P95 exceeds 8s. Answer labelled as 'quick mode' — quality degrades gracefully.

K. Human-in-the-Loop (HITL) Design

HITL is a first-class PM feature, not an afterthought

Competitive intelligence informs pricing decisions, roadmap priorities, and board presentations. Wrong information has real consequences. HITL is designed to be automatic, specific, and auditable.

When HITL triggers automatically

- Faithfulness score < 0.85 — hallucination guard flags the answer for review
- High-stakes keywords detected — queries containing 'pricing', 'acquisition', 'lawsuit', or 'partnership' trigger mandatory review
- Stale data — if the best available chunk is > 14 days old, a human verifies currency before delivery
- User explicitly requests verification — "Double-check this before I use it in the board deck"

HITL analyst workflow

- Analyst sees: the question, the generated answer, each cited chunk with its source URL and date, and the faithfulness score that triggered review
- Actions available: Approve as-is / Edit + approve / Reject and regenerate / Flag source as unreliable

- SLA: Urgent (board meeting context) = 2-hour SLA. Standard = 24-hour SLA. User is notified of expected delivery time immediately.
- Feedback loop: All HITL corrections are logged and used monthly to fine-tune the hallucination guard classifier. The system learns from its own mistakes.

L. Success Metrics

Day 90 launch success checklist

Metric	Target	Why this metric
Task Success Rate (primary)	> 80%	North star. Did the PM get something actionable? Measured by thumbs-up or no follow-up within 60s.
Hallucination rate	< 5%	Nightly RAGAS eval on 200-question golden test set. Alert if exceeded for 3 consecutive days.
P95 latency	< 5s	Users are in meetings. Latency SLA is in enterprise contracts.
DAU / MAU ratio	> 60%	Sticky daily usage means the product is in the workflow, not a monthly experiment.
Manual research time saved	> 50%	Surveyed at Day 30 and Day 90. The business outcome that justifies the product cost.
NPS at Day 90	> 40	Confirms product-market fit in the competitive intelligence category.
Cost per query	< \$0.03	Tracked per tenant in real time. Triggers optimisation sprint if exceeded.

M. Competitor Set Definition (NEW)

Why this section was missing from the original PRD

The Competitor Selector agent needs decision logic. Without defining how competitors are chosen, the agent has no rules — it is an engine without a steering wheel.

How the competitor set is defined

- User-defined list — PM manually adds known competitors at setup (always available)
- Industry category matching — system suggests competitors from Crunchbase by product category and funding stage
- Automatic discovery — agent monitors which companies appear repeatedly in news alongside the user's product name and flags them as potential additions
- Human approval required — automatic suggestions are never added without PM confirmation (HITL for competitor set curation)

N. Privacy & Legal Considerations (NEW)

Scraping and data use

- Respect robots.txt — the crawler checks and obeys robots.txt before accessing any page
- Public data only — no login-gated content, no personal data, no internal documents
- No EU personal data — if scraping EU-hosted sources, GDPR compliance is enforced: no storage of identifiable personal information
- Rate limiting — crawlers are rate-limited to avoid service disruption to target sites
- ToS review — high-risk sources (LinkedIn, Glassdoor) are reviewed for scraping permissions before inclusion

O. Rollout & Phasing Plan (NEW)

Why a phasing plan is required in a PM PRD

A PRD without a phasing plan is a design document, not a product plan. Each phase has a specific hypothesis to validate before moving to the next.

Phase	What ships	Hypothesis to validate	Exit criteria
v1 (Weeks 1–4)	Manual competitor list + weekly reports	PMs will pay for structured, cited reports they don't have to write themselves	5 beta users, Task Success Rate > 70%
v2 (Weeks 5–12)	Automated crawling + daily alerts for changes	Real-time alerts on competitor changes are more valuable than weekly summaries	DAU/MAU > 40%, < 20% churn in first 30 days
v3 (Weeks 13+)	Autonomous agent + query interface + Slack integration	PMs will use conversational Q&A as their primary research workflow, replacing manual search	Task Success Rate > 80%, NPS > 40 at Day 90

P. Key Innovation

What makes this different from a dashboard

Quarry AI acts as an autonomous AI analyst. It does not show data — it generates strategic recommendations with cited sources. It combines Agent + RAG + real-time intelligence in a way that no static dashboard or manual research process can match.

- Autonomous — continuously monitors competitors without requiring any manual action
- Grounded — every claim is traceable to a source URL and timestamp
- Fast — answers in under 5 seconds, suitable for use in live meetings
- Trustworthy — HITL gates prevent unverified answers from reaching high-stakes decisions
- Improving — HITL corrections feed back into the model; the system gets better over time